Zewail City School of Science and Technology	Group: 4th Year
School of Computational Sciences and	Course Name: Speech Recognition
Artificial Intelligence	Course Code: DSAI 456
Date: Jan 10, 2026	Time Allowed: 1.5 hours
Total Marks: 40	No. of Pages: 6

Answer the questions in the spaces provided on the question sheets.

GROUP A

Name: _____

Serial No.: _____

Question	Points	Score
1	3	
2	9	
3	8	
4	10	
5	10	
Total:	40	

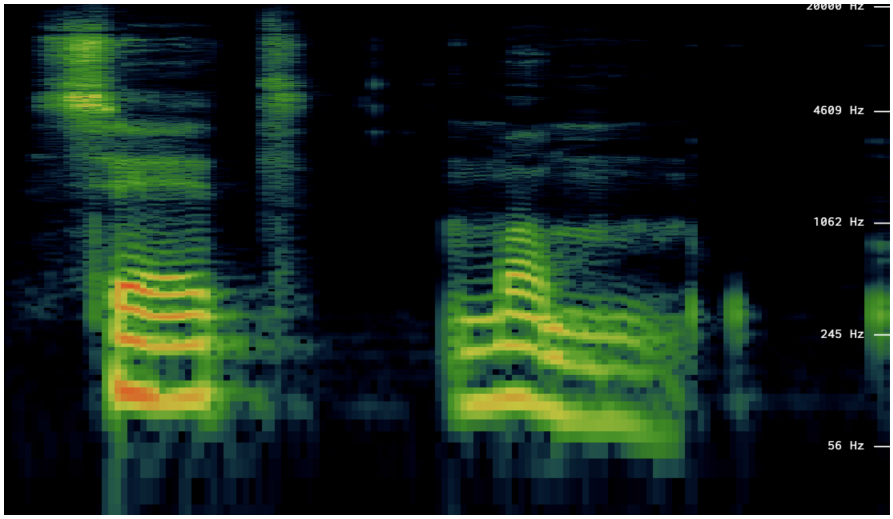
In **ALL** questions, you need to show **all the steps** required to obtain your answers.

Do not answer in this page!

Good luck!!!
Dr. Mohamed Ghalwash

Q1 (a) (1 point) How to compute the frequency of a signal that is not regular (aperiodic), i.e. the signal does not repeat at a fixed interval?

(b) (2 points) Fricative sounds are consonant sounds made by forcing air through a narrow channel in the vocal tract, creating a hissing or buzzing friction such as ‘f’, ‘v’, ‘s’, ‘sh’. Nasal murmur sounds refer to the specific low-frequency acoustic energy from nasal resonance during speech (like in ‘m’, ‘n’). Given the following frequency spectrogram, what is the most likely spoken words? Justify your answer.



A. MAM NAH
B. SAT MOM
C. Nah SAP
D. SAP SAT

Q2 Given a GMM

$$x \sim 0.3 * \mathcal{N}_1(3, 1) + 0.7 * \mathcal{N}_2(5, 0.5)$$

where

$$\mathcal{N}(x \mid \mu, \sigma^2) \;=\; \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

(a) (2 points) Plot the mixture of Gaussian.

(b) (4 points) Which component does the point 4 mostly generated from? Justify your answer by computing the responsibility of the point to each component.

A. Component 1

B. Component 2

(c) (3 points) Derive the closed-form solution for μ_k.

Q3 Consider a HMM used to identify a sequence of spoken phonemes based on detected acoustic energy. Assume there are two hidden states: H (High-frequency phoneme, e.g., 's') and L (Low-frequency phoneme, e.g., 'a'). The observations are: "hiss" (High-frequency noise detected) and "hum" (Low-frequency resonance detected). Assuming both states are equally probable to start with. Here are the transition A and emission matrices B.

$$A = \begin{bmatrix} P(H|H) & P(L|H) \\ P(H|L) & P(L|L) \end{bmatrix} = \begin{bmatrix} 0.6 & 0.4 \\ 0.2 & 0.8 \end{bmatrix}$$

$$B = \begin{bmatrix} P(\text{“hiss”}|H) & P(\text{“hum”}|H) \\ P(\text{“hiss”}|L) & P(\text{“hum”}|L) \end{bmatrix} = \begin{bmatrix} 0.7 & 0.3 \\ 0.2 & 0.8 \end{bmatrix}$$

(a) (3 points) Find the most likely sequence of hidden states given the observed sequence of acoustic features:

$$Y = (\text{“hiss”}, \text{“hum”})$$

Use the back of this page to solve this bullet

(b) (2 points) What is the objective function of the learning problem?

(c) (3 points) You are designing a system for a "Silent Speech" interface. The system uses a high-speed camera to track the vertical opening of the mouth (in millimeters) while a person speaks, but there is no audio. You need to use a HMM to determine if the person is saying a Vowel or a Plosive consonant (like 'P' or 'B').

i. What coud be the observations states?

ii. What could be the hidden states?

Q4 Assume a speech model has a vocabulary of $\Sigma = \{C, A, T, L, S, \epsilon\}$. For an audio clip of the word **'CAT'**, the RNN produces the following probability matrix Y for $T = 4$ time steps:

$$Y = \left[\begin{array}{c|cccccc} t & P(C) & P(A) & P(T) & P(L) & P(S) & P(\epsilon) \\ \hline 1 & 0.60 & 0.05 & 0.05 & 0.05 & 0.05 & 0.20 \\ 2 & 0.05 & 0.70 & 0.05 & 0.05 & 0.05 & 0.10 \\ 3 & 0.05 & 0.05 & 0.10 & 0.05 & 0.05 & 0.70 \\ 4 & 0.05 & 0.05 & 0.80 & 0.05 & 0.05 & 0.00 \end{array}\right]$$

(a) (3 points) What is the probability that the model predicts the incorrect alignment (C, A, A, T)?

(b) (3 points) Why did the model place a high probability (0.70) on ϵ at $t = 3$?

(c) (4 points) Apply the CTC Forward algorithm to compute the total probability of all paths that have finished the word "CAT" by time step 4.

Q5 (a) For the EnCodec model we studied in the lecture

i. (2 points) We spoke about one limitation in EnCodec, what was that?

(b) For the VALL-E model we studied in the lecture

i. (2 points) Explain the mechanism that enables zero-shot text-to-speech (TTS) synthesis. In particular, describe how the model generalizes to unseen speakers using only a brief acoustic prompt.

ii. (2 points) The architecture utilizes a Conditional Codec Language Model featuring both Autoregressive (AR) and Non-Autoregressive (NAR) Transformer decoders. Why is the first acoustic layer modeled using an AR decoder, while the subsequent layers utilize NAR decoders? Explain the structural or computational necessity for this distinction.

iii. (2 points) How many NAR decoder layers are implemented in this specific architecture

iv. (2 points) If the number of quantizers in RVQ is changed, does it necessitate a structural change in the Transformer blocks themselves? Justify your answer.